

Journey of GCD

Input file: **standard input**
Output file: **standard output**
Time limit: 3 seconds
Memory limit: 1024 megabytes

In the magical kingdom of Numerian, there are n cities, each with its own magical power level a_i . The king has built bidirectional roads between every pair of cities. For the road connecting city u and city v , its Guardian Charm Degree (GCD) equals $\gcd(a_u, a_v)$, which is the greatest common divisor of their magical power levels.

For a traveler moving from city x to y , the Magical Immunity Number (MIN) of the journey is the minimum GCD among all roads taken. The royal advisors consider a journey “well-immune” if and only if its MIN is maximized.

You are tasked with answering q queries from the king. For each query (x, y) , output the maximum possible MIN among all paths from city x to city y .

In particular, if $x = y$, the traveler never leaves the city, so the MIN is simply the city’s own magical power a_x .

Input

The first line contains two integers n and q ($1 \leq n, q \leq 10^6$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^6$), representing the magical power of each city.

Each of the next q lines contains two integers x and y ($1 \leq x, y \leq n$), representing a query.

Output

For each query, output a single integer representing the maximum possible MIN.

Example

standard input	standard output
6 3	7
6 15 22 33 35 63	3
5 6	15
3 5	
2 2	

Note

For the sample, the magical power levels are $a = [6, 15, 22, 33, 35, 63]$.

Query 1: The optimal path is $5 \rightarrow 6$, and the answer is $\gcd(35, 63) = 7$.

Query 2: The optimal path is $3 \rightarrow 4 \rightarrow 6 \rightarrow 5$. The edge values are $\gcd(22, 33) = 11$, $\gcd(33, 63) = 3$, and $\gcd(63, 35) = 7$, so the answer is $\min(11, 3, 7) = 3$.

Query 3: Since $x = y$, the traveler never leaves the city, so the answer is $a_2 = 15$.